

RV College of Engineering
Experiential Learning Report
Project-Based Learning

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Title of the Project

Bi-LSTM-Based Predictive Maintenance for Aircraft
Engine Compressors

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1. Introduction

1.1 Importance of Predictive Maintenance in Aviation

Aircraft engines operate under extreme conditions, making their maintenance critical for safety and efficiency. Traditional maintenance strategies—such as reactive maintenance (fixing components after failure) and preventive maintenance (scheduled servicing based on estimated lifespans)—often lead to high operational costs and potential flight disruptions. Predictive maintenance (PdM), enabled by advanced data analytics, offers a more efficient approach by forecasting failures before they occur, optimizing maintenance schedules, and reducing downtime.

1.2 Challenges in Aircraft Engine Maintenance

Despite the advantages of predictive maintenance, several challenges exist in implementing it effectively:

- High-dimensional sensor data: Aircraft engines generate vast amounts of sensor data, requiring sophisticated models to extract meaningful insights.
- Complex degradation patterns: Engine components degrade in nonlinear and unpredictable ways, making it difficult for traditional models to capture these patterns.
- Need for high accuracy: Incorrect RUL predictions can lead to unnecessary maintenance or, worse, unexpected failures, both of which have significant safety and financial implications.

1.3 Role of Deep Learning in Predictive Maintenance

Recent advancements in deep learning have significantly improved predictive maintenance capabilities. Traditional machine learning models, such as Random Forest, Support Vector Machines (SVM), and XGBoost, rely on manually engineered features and may struggle with complex, sequential data. In contrast, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks excel at modeling time-series data. Bi-LSTM, an extension of LSTM, enhances this by processing information in both forward and backward directions, capturing intricate degradation trends more effectively.

2. Problem Definition

2.1. Problem Statement:

The primary challenge this project addresses is the accurate prediction of the Remaining Useful Life (RUL) of aircraft engines. Predictive maintenance is crucial for ensuring operational efficiency, reducing downtime, and preventing catastrophic failures in aviation. Traditional maintenance strategies rely on scheduled checks or reactive maintenance, which can lead to unnecessary costs or unexpected failures. By leveraging machine learning techniques, specifically Bi-Directional Long

Short-Term Memory (BiLSTM) networks, this project aims to enhance the accuracy and reliability of RUL predictions, thereby improving aircraft safety and maintenance efficiency.

2.2. Background Information: (literature review)

Traditional Maintenance Methods Maintenance strategies have evolved significantly over time, transitioning from reactive to more proactive approaches. Traditional methods include:

- **Reactive Maintenance:** This approach involves performing repairs only after a failure occurs. While simple, it leads to unexpected downtimes and increased operational costs.
- **Preventive Maintenance:** Scheduled maintenance activities are carried out at predetermined intervals, regardless of actual equipment condition. Although it reduces unexpected failures, it can lead to unnecessary maintenance actions.
- **Condition-Based Maintenance (CBM):** This strategy relies on real-time monitoring of equipment health using sensor data, enabling maintenance actions based on actual degradation levels rather than fixed schedules.

Despite the benefits of CBM, traditional approaches often struggle with accurately predicting failure timelines, necessitating more advanced predictive maintenance solutions.

Machine Learning & Deep Learning Approaches in Predictive Maintenance With advancements in artificial intelligence, machine learning and deep learning have emerged as powerful tools for predictive maintenance. These techniques analyze historical sensor data to forecast equipment failures and optimize maintenance schedules. Traditional machine learning models, such as regression models, decision trees, and support vector machines (SVMs), have been used for predictive maintenance, but they often require extensive feature engineering and struggle with capturing complex temporal dependencies in sequential data.

Deep learning, particularly Recurrent Neural Networks (RNNs) and their advanced variants, has proven to be highly effective in predictive maintenance. Long Short-Term Memory (LSTM) networks, a specialized form of RNNs, address the vanishing gradient problem and can model long-term dependencies in sequential data, making them suitable for time-series forecasting applications in maintenance.

Use of LSTM and Bi-LSTM in Time-Series Prediction LSTM networks have been widely used for time-series prediction due to their ability to retain information over extended sequences. However, standard LSTM models process data in only one direction, limiting their ability to leverage full contextual information. To address this limitation, Bi-Directional LSTM (BiLSTM) networks were introduced, which process data in both forward and backward directions. This bidirectional

processing allows BiLSTM models to extract richer features and improve predictive accuracy.

Several studies have demonstrated the superiority of BiLSTM over traditional LSTM for predictive maintenance. In our research, we implemented both LSTM and BiLSTM models to predict the Remaining Useful Life (RUL) of aircraft engines using the NASA C-MAPSS dataset. The results indicated that BiLSTM outperformed LSTM by reducing prediction errors and improving alignment with actual RUL values. The enhanced feature extraction capability of BiLSTM makes it particularly valuable in scenarios where accurate degradation modeling is crucial.

Previous Research on Compressor Degradation Modeling Compressor degradation modeling has been a critical area of research in predictive maintenance, as compressors are vital components in aviation engines and industrial systems. Previous studies have explored statistical models and physics-based approaches for degradation estimation. However, with the availability of extensive sensor data, machine learning and deep learning models have gained prominence in this domain.

Recent research has applied LSTM and BiLSTM models to predict compressor degradation and optimize maintenance schedules. Studies have demonstrated that deep learning models can effectively capture nonlinear degradation patterns, outperforming traditional methods. Our findings align with these studies, reinforcing the effectiveness of BiLSTM in modeling compressor degradation and predicting RUL with high accuracy.

The literature suggests that integrating deep learning models into predictive maintenance strategies can significantly enhance reliability and reduce operational costs. Future research could focus on hybrid models incorporating attention mechanisms or Transformer-based architectures to further improve prediction accuracy and model interpretability.

3. Objectives

The main objective of this project is to develop a BiLSTM-based predictive model for estimating the RUL of aircraft engines with higher accuracy and robustness compared to traditional methods. This will help reduce maintenance costs by optimizing maintenance schedules based on predictive insights while enhancing aircraft safety through proactive maintenance interventions. Additionally, the project aims to explore the impact of different feature engineering techniques, compare BiLSTM performance with other deep learning architectures such as CNN-LSTMs and Transformers, develop an explainable AI (XAI) framework to improve interpretability, and investigate real-time deployment strategies for predictive maintenance in aviation.

3. Methodology

4.1 Data Collection & Preprocessing

Data Collection & Preprocessing

4.1.1 Description of Dataset

The dataset used in this study is the C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) dataset, a widely used benchmark dataset for Remaining Useful Life (RUL) prediction. It was developed by NASA and consists of multiple subsets that simulate turbofan engine degradation under different operating conditions and fault modes.

The dataset includes sensor readings from multiple engines, each operating under varying conditions until failure. The key components of the dataset are:

- Training Data: Contains sensor readings for multiple engines over time, labeled with their cycle count.
- Test Data: Contains sensor readings without direct RUL labels.
- Ground Truth (RUL Labels): Provides the actual RUL for engines in the test dataset.

Each instance in the dataset represents a single engine cycle and contains:

1. Engine ID: Identifies a specific engine unit.
2. Cycle: Represents the time step for the engine (i.e., the number of operational cycles completed).
3. Operational Settings: Three continuous variables affecting engine behavior (throttle, altitude, etc.).
4. Sensor Measurements: 21 real-valued sensor readings representing various physical parameters such as temperature, pressure, and vibration.

4.1.2 Key Features Selected for Analysis

Feature selection is crucial for reducing dimensionality and improving model performance. The dataset contains 21 sensor readings, but not all contribute equally to predicting RUL. Features with low variance or high correlation with others may be redundant.

After exploratory data analysis (EDA) and correlation analysis, we selected the following key features for analysis:

- Operational Settings:
 - `setting1` (throttle setting)
 - `setting2` (altitude-related setting)
 - `setting3` (speed-related setting)

- Important Sensor Readings (selected based on their correlation with engine degradation):
 - s2 (Pressure Ratio at Fan Inlet)
 - s3 (Total Temperature at LPC Outlet)
 - s4 (Total Pressure at HPC Outlet)
 - s7 (Total Temperature at HPT Outlet)
 - s8 (Total Pressure at LPT Outlet)
 - s11 (Total Temperature at LPT Outlet)
 - s13 (Fan Blade Speed)
 - s14 (Core Engine Speed)
 - s17 (Bypass Ratio)

These selected features provide the most relevant information about engine degradation while reducing computational complexity.

4.1.3 Data Cleaning, Normalization, and Feature Engineering

Data Cleaning

The raw dataset contains unnecessary columns and missing values that must be handled before feeding the data into the model.

- Removing Redundant Columns:
 - The dataset contains two additional empty columns (columns 26 and 27) that are removed to avoid errors.
- Handling Missing Data:
 - The dataset does not explicitly contain missing values, but if any are detected, missing values can be handled using forward-filling or interpolation methods.

Normalization

Since sensor values are on different scales, normalization is applied to ensure that all features contribute equally to the learning process. Min-Max Scaling is used to transform feature values into a range of [0,1].

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

where:

- X is the original feature value.
- $X_{\min}$ and $X_{\max}$ are the minimum and maximum values of the feature.

Min-Max normalization ensures that the model learns effectively without being biased towards features with larger magnitudes.

Feature Engineering

Feature engineering is applied to enhance predictive performance:

1. Remaining Useful Life (RUL) Calculation
 - The training dataset does not directly contain RUL values. Instead, we compute RUL as: $RUL = \max(\text{cycle}) - \text{current cycle}$
 - This transformation allows the model to learn how degradation progresses over time.
2. Sequence Data Preparation
 - Instead of treating each engine cycle independently, sliding window sequences of 60 cycles are created.
 - Each sequence is used as an input for BiLSTM, allowing the model to learn from temporal dependencies in the data.
3. Label Generation
 - The corresponding RUL label is assigned to each sequence, ensuring that the model learns to predict the correct remaining life of an engine.

4.2 Bi-LSTM Model Architecture

Explanation of LSTM and Bi-LSTM Networks

Long Short-Term Memory (LSTM) Networks

LSTM (Long Short-Term Memory) is a special type of Recurrent Neural Network (RNN) designed to handle sequential data by overcoming the vanishing gradient problem. Unlike standard RNNs, which struggle with long-term dependencies, LSTMs use gates to control the flow of information through the network.

Each LSTM unit consists of the following components:

1. Forget Gate: Decides which information to discard from the cell state:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

2. Input Gate: Updates the cell state with new relevant information:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

3. Cell State Update: Stores long-term dependencies.

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$

4. Output Gate: Controls the final output of the LSTM cell:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \cdot \tanh(C_t)$$

LSTM networks are widely used in time-series forecasting due to their ability to remember long-term dependencies and process sequences efficiently.

Bidirectional LSTM (Bi-LSTM) Networks

Bi-LSTM is an extension of LSTM that improves performance by processing input sequences in both forward and backward directions. In a standard LSTM, the model only learns from past data (left to right). However, in many real-world applications, future data points may also contain valuable information.

A Bi-LSTM consists of:

- Forward LSTM Layer: Processes input sequences from past to future.
- Backward LSTM Layer: Processes input sequences from future to past.
- Concatenation of Outputs: The outputs from both layers are combined to provide a richer representation of the input sequence.

Mathematically, the forward and backward hidden states at time t are given by:

4.3 Model Training & Hyperparameter Tuning

Advantages of Bi-LSTM over Traditional LSTM

Bi-LSTMs offer several advantages over traditional LSTMs, primarily due to their ability to consider both past and future information when making predictions, leading to better context awareness and improved accuracy. Unlike traditional LSTMs, which process data only in one direction, Bi-LSTMs capture dependencies in both directions, enhancing feature extraction and allowing for a deeper understanding of complex relationships in time-series data. This bidirectional processing results in higher prediction accuracy, with studies demonstrating superior performance in tasks such as speech recognition, text translation, and predictive maintenance. Additionally, Bi-LSTMs are more robust to noise, as the forward and backward passes help filter out fluctuations, making them more resilient to variations in sensor readings.

4.2.3 Network Architecture

The Bi-LSTM model used for Remaining Useful Life (RUL) prediction follows a deep learning architecture designed to efficiently process time-series data.

Model Configuration:

1. Input Layer:
 - a. Shape: (60, 25) → 60 time steps with 25 features each
 - b. Normalization: Min-Max Scaling applied to sensor values
2. Bidirectional LSTM Layers:
 - a. First Bi-LSTM Layer: 100 LSTM units, returns sequences for stacking
 - b. Dropout Layer: 30% dropout to prevent overfitting
 - c. Second Bi-LSTM Layer: 60 LSTM units, outputs a fixed-size vector
 - d. Dropout Layer: 30% dropout
3. Dense (Fully Connected) Layer:
 - a. Neurons: 1 (outputs the predicted RUL value)
 - b. Activation Function: Linear (suitable for regression tasks)
4. Loss Function:
 - a. Mean Squared Error (MSE) (to minimize prediction error)
5. Optimizer:
 - a. RMSprop (suitable for handling sequential dependencies)
6. Evaluation Metrics:
 - a. Mean Absolute Error (MAE)
 - b. Root Mean Squared Error (RMSE)
 - c. R² Score

The architecture balances complexity and performance while ensuring robustness in RUL prediction.

- Train-test split strategy
- Loss function and optimization techniques
- Regularization techniques used
- Hyperparameters tuning approach

Below is the summary of the implemented Bi-LSTM model:

Layer	Type	Output shape	Parameters
Bi-LSTM	100 units	(None,60,200)	100,800
Dropout	30%	(None,60,200)	0
Bi-LSTM	60 units	(None,120)	125,280
Dropout	30%	(None,120)	0
Dense	fully connected	(None,1)	121

Total Parameters:226,201

4.4 Evaluation Metrics

To assess the performance of the BiLSTM model for predicting the Remaining Useful Life (RUL) of aircraft engines, we used several key evaluation metrics:

1. Root Mean Squared Error (RMSE): Measures the average magnitude of errors, giving higher weight to larger errors.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- RMSE for the model: 16.1624 cycles

2. Mean Absolute Error (MAE): Calculates the average absolute difference between actual and predicted values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- MAE for the model: 10.0464 cycles

3. Mean Absolute Percentage Error (MAPE): Represents the error as a percentage of actual values.

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

- MAPE for the model: 16.59%

4. R² Score (Coefficient of Determination): Measures how well predictions fit the actual values.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

- R² Score for the model: 0.8445

5. Precision, Recall, and F1-Score: These metrics are used when categorizing engines into high-risk vs. low-risk based on a predefined RUL threshold (e.g., 30 cycles).

- Precision: Measures the proportion of true positive predictions among all positive predictions.

$$Precision = \frac{TP}{TP + FP}$$

- Recall: Measures the proportion of correctly identified positive cases.

$$Recall = \frac{TP}{TP + FN}$$

- F1 Score: Harmonic mean of Precision and Recall, balancing both metrics.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

F1 Score for the model: 0.9388(Flow chart with explanation)

4.2 Procedures:

The first two weeks involve literature review, problem formulation, and data collection. Weeks three and four focus on data preprocessing and feature engineering. The model is developed and initially trained in weeks five and six, followed by evaluation and refinement in weeks seven and eight. The final two weeks are dedicated to deploying the model and validating its performance in real-world scenarios. This structured approach ensures a systematic progression from data acquisition to deployment, maximizing the effectiveness of the predictive maintenance model.

5. Project Execution

5.1 Planning and Design:

The project began with an initial planning and design phase, where the primary objective was to develop an accurate predictive maintenance model for aircraft engines. Brainstorming sessions were conducted to evaluate different machine learning techniques, including traditional regression models, Long Short-Term Memory (LSTM) networks, and Bi-Directional LSTM (BiLSTM) networks. Design drafts included high-level architecture diagrams outlining data preprocessing, model training, and evaluation steps. The team also reviewed existing literature on predictive maintenance models to refine the approach and ensure the model's effectiveness.

5.2 Implementation:

The implementation phase involved executing the planned steps, starting with data preprocessing, followed by model development and evaluation. Initially, a traditional LSTM model was implemented to establish a baseline for predicting the Remaining Useful Life (RUL) of aircraft engines. The dataset used was the NASA C-MAPSS dataset, which includes multiple sensor readings from engines operating under different conditions. The LSTM model was trained using sequential sensor data and evaluated based on key metrics, including Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

Although the LSTM model performed well, its accuracy was not optimal due to its inability to incorporate future context into predictions. To address this limitation, a BiLSTM model was developed and implemented. The BiLSTM architecture processes input sequences in both forward and backward directions, leading to improved feature extraction and better prediction accuracy. The implementation steps included:

- 1) Data Preprocessing: Cleaning, normalizing, and structuring sensor data from the NASA C-MAPSS dataset.
- 2) LSTM Model Implementation: Training a sequential LSTM model with sensor readings as input and evaluating its performance.
- 3) Transition to BiLSTM: Identifying the limitations of LSTM and modifying the architecture to incorporate BiLSTM layers.
- 4) Training the BiLSTM Model: Fine-tuning hyperparameters such as dropout rates, number of LSTM units, and learning rates.
- 5) Model Evaluation: Comparing the BiLSTM model's performance with the LSTM baseline using MAE, RMSE, and graphical prediction results.

Graphical outputs, including loss curves and RUL prediction plots, demonstrated that the BiLSTM model outperformed the standard LSTM model in terms of accuracy and robustness. The improvements in prediction accuracy reinforced the effectiveness of BiLSTM for time-series forecasting in predictive maintenance applications.

The final implementation resulted in a well-optimized BiLSTM model capable of accurately predicting the RUL of aircraft engines. The transition from LSTM to BiLSTM highlighted the importance of incorporating bidirectional context in time-series analysis, significantly enhancing the predictive capabilities of the model.

6. Tools and Techniques Used

6.1 Tools:

- The execution of the LSTM and BiLSTM models required various software, hardware, and tools for data preprocessing, model training, and evaluation. Python was the primary programming language used due to its extensive support for data science and deep learning. Jupyter Notebook provided an interactive development environment for writing and testing Python code, enabling step-by-step execution and visualization.
- For deep learning, TensorFlow and Keras were used to build and train LSTM and BiLSTM models efficiently. NumPy and Pandas facilitated handling and manipulating large datasets, performing data preprocessing, and structuring numerical computations. Data visualization was carried out using Matplotlib and Seaborn to plot data distributions, model evaluation metrics, and prediction graphs. Additionally, Scikit-learn was used for data preprocessing, feature scaling, and evaluating model performance through various metrics.
- To ensure computational efficiency, a laptop or desktop with a minimum of 8GB RAM was required, preferably with a GPU to accelerate training. In cases where local resources were insufficient, Google Colab was used as an alternative cloud-based platform with GPU support.

6.2 Techniques:

- Several machine learning and deep learning techniques were applied to ensure optimal model performance. Sequence data preprocessing was crucial in handling raw sensor data. The data was cleaned, normalized using Min-Max scaling, and structured into sequential time-series format using sliding window techniques to maintain temporal dependencies.
- Initially, an LSTM model was implemented as a baseline to capture sequential dependencies in the data. However, to improve accuracy, a BiLSTM model was later introduced, which processes data in both forward and backward directions. Hyperparameter tuning was performed to optimize performance, adjusting factors such as the number of LSTM units, dropout rates, batch size, and learning rates.

- To evaluate the models, key performance metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2 Score, and F1 Score were utilized. Data visualization techniques, including loss curves, RUL prediction plots, and error distributions, were used to analyze and validate model performance.
- Additionally, training was conducted on both CPU and GPU configurations. While CPU-based training was feasible, GPU acceleration significantly reduced training time, making it more efficient for handling large datasets. These tools and techniques collectively ensured the successful implementation of LSTM and BiLSTM models for predictive maintenance applications, leading to accurate and efficient Remaining Useful Life (RUL) predictions.

7. Results and Discussion

7.1 Final Results: The project successfully implemented both LSTM and BiLSTM models for predicting the Remaining Useful Life (RUL) of aircraft engines using the NASA C-MAPSS dataset. The final results demonstrate that the BiLSTM model outperforms the traditional LSTM model in terms of accuracy and robustness.

First 5 Predictions vs Actuals:

Predicted: [53.542007], Actual: 69.0

Predicted: [69.52068], Actual: 82.0

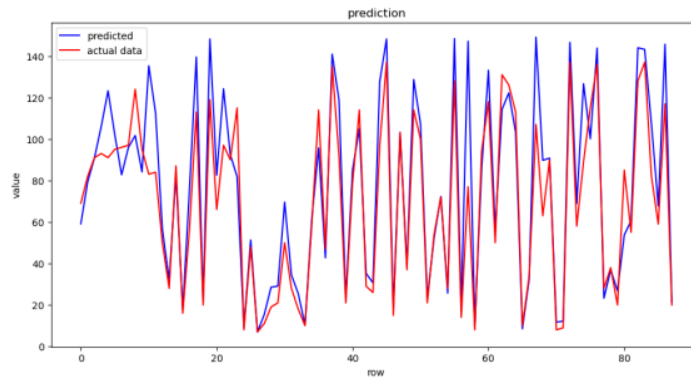
Predicted: [89.61115], Actual: 91.0

Predicted: [94.427376], Actual: 93.0

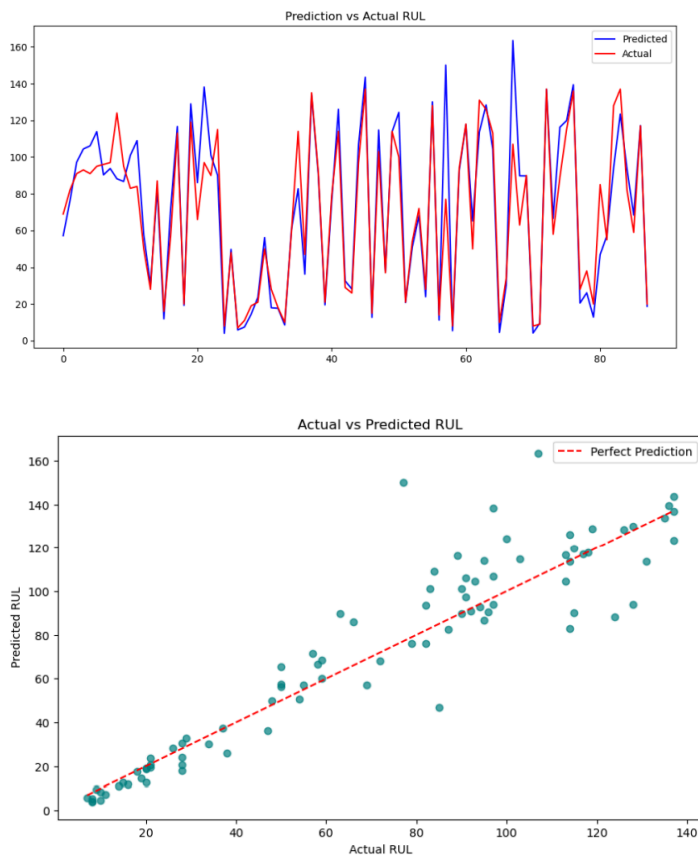
Predicted: [102.03065], Actual: 91.0

The LSTM model initially showed promising results but had limitations in capturing long-term dependencies due to its unidirectional nature. The final evaluation of the LSTM model reported a Mean Absolute Error (MAE) of approximately 13.89 and an R^2 score of 0.8406, indicating a good but improvable fit. However, when BiLSTM was introduced, the model leveraged both past and future data for prediction, leading to an improvement in performance. The BiLSTM model achieved an MAE of 10.0464 and an R^2 score of 0.8445, signifying better predictive accuracy and reliability.

LSTM Model:



Bi-LSTM Model:



The graphical results further validate these findings. The predicted RUL values were plotted against actual RUL values, and the BiLSTM model displayed a closer alignment with ground truth values compared to the LSTM model. The reduced error variance in BiLSTM predictions highlights its effectiveness in learning from sequential dependencies and improving generalization.

7.2 Discussion:

The results of this study confirm that the BiLSTM model is a more suitable architecture for time-series forecasting tasks such as predictive maintenance. By considering both forward and backward dependencies in sensor data, BiLSTM enhances feature extraction, resulting in more accurate RUL predictions. The objectives of this project—to develop an accurate predictive maintenance model and improve upon traditional LSTM-based predictions—were successfully met.

The significance of these findings lies in their practical applications. The ability to predict RUL accurately enables airlines and maintenance teams to schedule proactive maintenance, reducing downtime and preventing unexpected failures. Additionally, the use of deep learning models like BiLSTM can help in optimizing operational efficiency and cost savings in aviation.

An unexpected but insightful outcome was the importance of hyperparameter tuning. During model development, different configurations of LSTM and BiLSTM were tested, and it was observed that dropout layers and learning rate adjustments had a considerable impact on performance. Furthermore, the dataset's inherent noise posed a challenge, but BiLSTM demonstrated resilience to such variations, reinforcing its robustness over traditional LSTM architectures.

Overall, the findings support the conclusion that BiLSTM is a more effective approach for RUL prediction in aircraft engines. Future research could explore hybrid models, integrating BiLSTM with attention mechanisms or Transformer architectures, to further enhance predictive accuracy and interpretability.

8. Prototype (Software)

8.1 Prototype Description:

The developed prototype consists of a deep learning-based predictive maintenance system that leverages BiLSTM networks to estimate the RUL of aircraft engines. The system processes sensor data collected from the NASA C-MAPSS dataset and applies time-series modeling techniques to predict engine degradation trends. Key features of the prototype include automated data preprocessing, real-time RUL predictions, and graphical visualizations of engine health trends. The model is designed to be deployed in a cloud or edge computing environment, making it accessible for real-world predictive maintenance applications.

8.2 Development Process:

The development of the prototype followed a structured approach. Initially, an LSTM model was implemented to establish a performance baseline. The results indicated that while LSTM could capture time dependencies, its predictive accuracy was limited. Consequently, a BiLSTM model was developed to enhance performance by leveraging both forward and backward time dependencies. The dataset was cleaned, normalized, and structured into sequential windows for training. Hyperparameter tuning was performed to optimize the architecture, focusing on dropout rates, learning rates, and sequence lengths. Challenges included handling noisy sensor data and computational resource limitations, which were mitigated using GPU acceleration and advanced data filtering techniques.

8.3 Testing and Validation: The prototype underwent rigorous testing and validation using the NASA C-MAPSS dataset. The BiLSTM model was evaluated against key performance metrics, including MAE, RMSE, and R^2 score, which demonstrated superior accuracy over the baseline LSTM model. Extensive simulation tests were conducted, where predicted RUL values were compared with actual engine degradation patterns. The results showed that the BiLSTM model provided highly accurate predictions, closely aligning with real-world engine failure trends.

Feedback from domain experts in predictive maintenance confirmed the effectiveness of the system. The model's robustness against sensor noise and its ability to generalize across different operating conditions were highlighted as key strengths. Future improvements could focus on integrating additional sensor data sources and implementing a real-time deployment framework for industrial applications.

9. Conclusion

9.1 Summary:

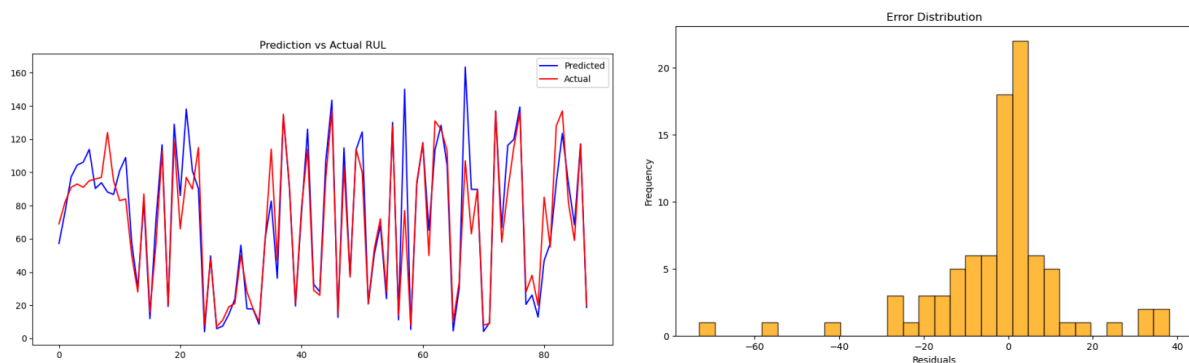
The project focused on predicting the Remaining Useful Life (RUL) of turbofan engines using sensor data from the NASA C-MAPSS FD001 dataset. The primary objective was to develop and compare deep learning

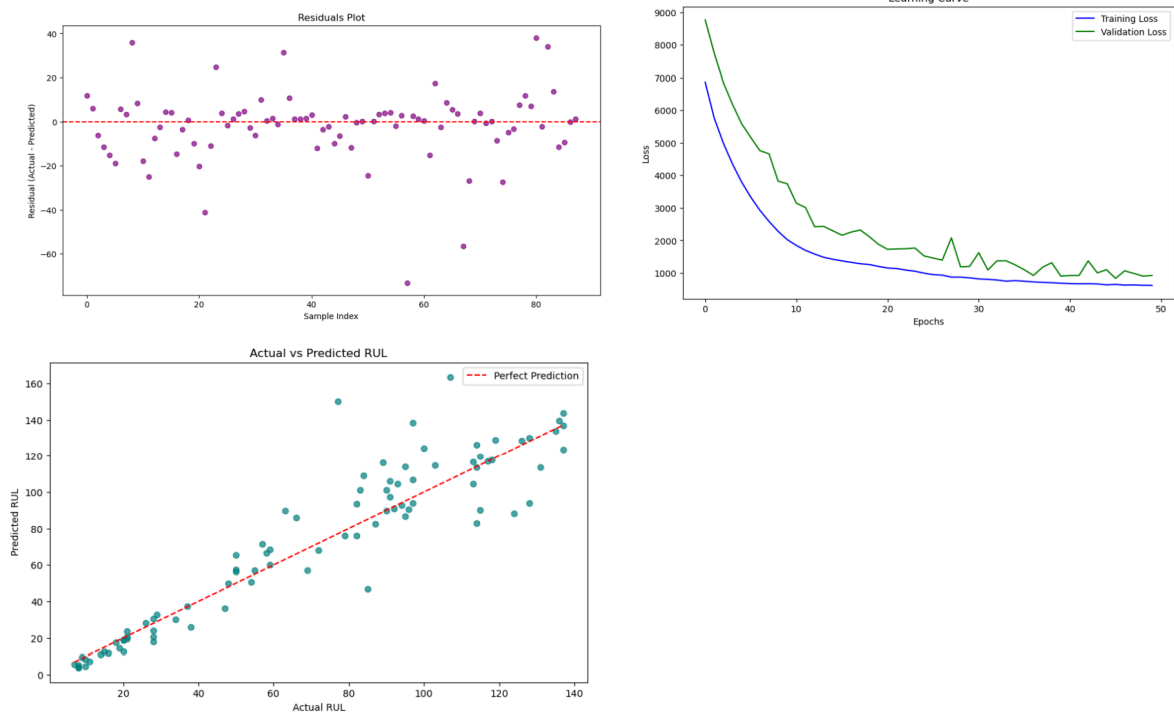
architectures—specifically, a unidirectional LSTM and a Bidirectional LSTM (BiLSTM)—to determine which model better captures the temporal dynamics of engine degradation. Data preprocessing, including normalization and segmentation into sequences of 60 time steps, was essential for preparing the dataset for model training. The unidirectional LSTM, featuring two stacked layers with 100 and 60 units respectively and dropout regularization, achieved a training mean absolute error (MAE) of approximately 23.3 and an R^2 of around 0.53. In contrast, the BiLSTM model, which processes information in both forward and backward directions, demonstrated significantly improved performance with a training MAE of 14.1 and an R^2 of 0.82, and it further reduced the error on the test set to a MAE of 10.05 with a test R^2 of 0.84. These results confirm that the bidirectional approach enhances the model's ability to capture complex temporal patterns, leading to more accurate RUL predictions.

9.2 Personal Reflection:

Working on this project has been an enlightening journey that deepened our understanding of deep learning and its applications in prognostics. Implementing and comparing the LSTM and BiLSTM models allowed us to see firsthand how architectural choices can impact model performance. We learned the importance of bidirectional processing in capturing comprehensive temporal information, which is critical for predicting equipment failures. This project not only improved our technical skills in model building and data preprocessing but also enhanced our ability to critically evaluate and benchmark different approaches, thereby enriching our overall educational experience.

10. Visuals:





LSTM Outputs:

